Answer Key — XII IP Preparatory-1

Section A (14 × 1 = 14 Marks)

Q	Answer
1	**b) RIGHT()**
2	**a) Uniform Resource Locator**
3	**b) MONTHNAME()**
4	**b) A small text file stored on the user's computer by a website**
5	**c) LTRIM()**
6	**b) 1**
7	**False**
8	**c) PAN**
9	**b) ORDER BY ... DESC**
10	**b) Chat**
11	**c) LCASE('DATA')**
12	**b) Mozilla Firefox**
13	**c) A is True but R is False.**
14	**c) A is True but R is False.**

Section B (4 × 2 = 8 Marks)

15. A. DELETE vs DROP

DELETE	DROP
Removes rows (tuples) from a table	Removes the entire table (structure + data)
Can use WHERE clause	Cannot use WHERE clause
Table structure remains	Table structure is deleted
Example: `DELETE FROM GROCERY WHERE Item_ID='G201';`	Example: `DROP TABLE GROCERY;`

OR

B. MAN (Metropolitan Area Network): A network that covers a city or a large campus, typically spanning up to 50 km. It connects multiple LANs within a city.
Example: Cable TV network in a city, or a network connecting all branches of a bank within a city.

WAN (Wide Area Network): A network that spans a very large geographical area, such as a country, continent, or the entire world. It connects multiple LANs and MANs.
Example: The Internet, or a network connecting offices of a company across different countries.

---

16. Outputs on GROCERY table

I. `SELECT SUM(Price), AVG(Price) FROM GROCERY;`

SUM(Price)	AVG(Price)
500	100.0000

II. `SELECT COUNT(*) FROM GROCERY WHERE Price > 50;`

COUNT(*)
3

---

17. Static Web Page vs Dynamic Web Page

Static Web Page	Dynamic Web Page
Content remains fixed and does not change unless the HTML file is manually edited	Content is generated dynamically and can change based on user input or database queries
Written using only HTML and CSS	Written using server-side languages like PHP, ASP.NET, JSP along with HTML
Faster to load	Slower to load compared to static pages
Example: A personal portfolio page, an informational "About Us" page	Example: An e-commerce website's product listing page, a social media feed

---

18. Green Valley School Network Scenario

I. The type of network formed within the computer lab is a **LAN (Local Area Network)**.

II. A **Switch** should be suggested to connect all computers within the lab.
Reason: A switch is a central device that connects multiple computers within a LAN and forwards data only to the intended destination device, making communication efficient and reducing collisions.

Section C (4 × 3 = 12 Marks)

19. A. Web Server & Web Hosting

Web Server: A computer program (or a computer) that stores, processes, and delivers web pages to clients (web browsers) upon request using HTTP/HTTPS protocol. It responds to client requests by sending the requested web content.

Web Hosting: A service that provides storage space on a server connected to the Internet, allowing individuals or organizations to make their websites accessible on the World Wide Web. The hosting company provides the server and related infrastructure.

Difference: A web server is the hardware/software that serves web pages, while web hosting is the service of providing space on a web server for a website.

OR

B. Browser Add-ons: Small software modules or extensions that add new features or functionality to a web browser (e.g., ad blockers, password managers).
Example: AdBlock Plus, Grammarly extension.

Plug-ins: Software components that add specific capabilities to a web browser, often to handle special content types like multimedia or documents.
Example: Adobe Flash Player (older), PDF viewer plug-in.

---

20. Online Shopping Scenario

I. These small files are called **cookies**.

II. Two commonly used web browsers: **Google Chrome** and **Mozilla Firefox** (also acceptable: Microsoft Edge, Safari).

III. A **web server** stores the website's files and processes requests from the user's browser. It delivers the web pages, handles login authentication, and manages the session data (via cookies) so that the website can remember the user's preferences and cart items.

---

21. Queries on PATIENT table

```
I.  SELECT UPPER(PName) FROM PATIENT WHERE Dept='Cardiology';

II. SELECT PNo, MONTHNAME(DOA) FROM PATIENT;

III.SELECT PName FROM PATIENT ORDER BY PName DESC;
```

---

22. Outputs on EMPLOYEE table

I. `SELECT LOWER(Name), LENGTH(Name) FROM EMPLOYEE WHERE ID = 2;`

LOWER(Name)	LENGTH(Name)
meera	5

II. `SELECT MONTH(DOB), DAY(DOB) FROM EMPLOYEE WHERE ID = 3;`

MONTH(DOB)	DAY(DOB)
11	5

III. `SELECT LEFT(Name, 2) FROM EMPLOYEE WHERE ID = 1;`

LEFT(Name, 2)
Ka

Section D (3 × 4 = 12 Marks)

23. Queries on BOOKS and AUTHORS tables

```
I.  SELECT B.Title, A.Author_Name, B.Price
    FROM BOOKS B, AUTHORS A
    WHERE B.Author_ID = A.Author_ID AND B.Genre = 'Fiction';

II. SELECT A.Country, COUNT(*) AS Total_Books
    FROM BOOKS B, AUTHORS A
    WHERE B.Author_ID = A.Author_ID
    GROUP BY A.Country;

III.SELECT B.Title, A.Author_Name
    FROM BOOKS B, AUTHORS A
    WHERE B.Author_ID = A.Author_ID
    AND B.Price > (SELECT AVG(Price) FROM BOOKS);

IV. SELECT B.Title, A.Author_Name, B.Price
    FROM BOOKS B, AUTHORS A
    WHERE B.Author_ID = A.Author_ID
    ORDER BY A.Author_Name ASC, B.Price DESC;
```

---

24. CareWell Hospital Network Design

I. Most suitable block for server

Diagnostics block should be chosen to install the server.
Reason: Diagnostics has the highest number of computers (120), so it is the largest block. Placing the server in the block with the maximum computers minimizes cabling cost and provides efficient access to the majority of users.

II. Cable layout

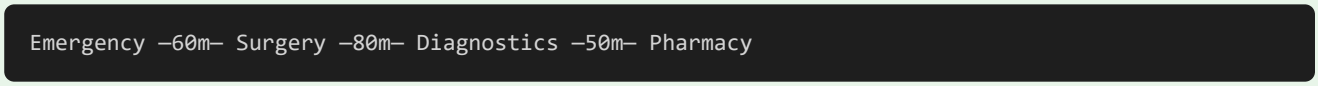
A suitable cable layout connects the blocks in a star or tree topology:



Since Diagnostics is the hub (most computers), it can act as the central node connecting to Emergency (via Surgery) and Pharmacy. Alternatively, a star topology with Diagnostics at the center connecting to all other blocks is suitable.

OR

Most cable-efficient layout (Minimum Spanning Tree):



Total cable required = 60 + 80 + 50 = 190 meters

Justification: This layout connects all four blocks using the minimum total cable length. Diagnostics is still the server location (most computers), and the chain avoids redundant connections.

Alternative (Star topology with Diagnostics at center):



Total cable = 100 + 80 + 50 = 230 meters. This is also acceptable if justified by placing the server at the block with the most computers.

III. Hardware device within each block

> A **Switch** should be used to connect all computers within each block.

IV. Most suitable network topology

> **Star Topology** is most suitable for connecting the blocks.
> **Justification:** In star topology, all blocks connect to a central hub/switch (placed in Diagnostics). This makes the network easy to manage, fault isolation is simple (if one cable fails, only that block is affected), and it is cost-effective for a campus with a central server location.

OR

> **Tree Topology** is most suitable for connecting the blocks in this campus.
>
> **Justification:**
>
> - Each block (Emergency, Surgery, Diagnostics, Pharmacy) forms its own **star network** with a switch connecting all computers within that block.
> - These block-level switches are then connected together hierarchically to a **main switch/server** (placed in Diagnostics, which has the most computers).
> - This hierarchical arrangement of multiple star networks is called **Tree Topology**.
> - Tree topology is scalable — new blocks can be added easily by connecting to the main switch.
> - Fault isolation is easy — if one block's switch fails, only that block is affected.
>
> **Note:** Within each block, **Star Topology** is used (computers connected to a central switch). Between blocks, **Tree Topology** is formed (block switches connected hierarchically).

---

25. Outputs on STUDENT and SCORES tables

I. `SELECT Name FROM STUDENT WHERE Class = '12' ORDER BY Name ASC;`

Name
Pooja
Riya
Sneha

II. `SELECT DISTINCT UPPER(Subject) FROM SCORES WHERE Score > 80;`

UPPER(Subject)
BIOLOGY
COMPUTER SCIENCE

MATHEMATICS

III. `SELECT STUDENT.Name, SCORES.Subject, SCORES.Score FROM STUDENT, SCORES WHERE STUDENT.Roll_No = SCORES.Roll_No;`

Name	Subject	Score
Riya	Mathematics	88
Riya	Physics	76
Arjun	Chemistry	90
Sneha	Mathematics	95
Vikram	Biology	82
Pooja	Computer Science	91
Arjun	Mathematics	79

IV. `SELECT STUDENT.Name, SUM(SCORES.Score) AS Total_Score FROM STUDENT, SCORES WHERE STUDENT.Roll_No = SCORES.Roll_No GROUP BY STUDENT.Name HAVING COUNT(SCORES.Subject) > 1;`

Name	Total_Score
Arjun	169
Riya	164

V. `SELECT Subject, AVG(Score) AS Average_Score FROM SCORES GROUP BY Subject;`

Subject	Average_Score
Biology	82.0000
Chemistry	90.0000
Computer Science	91.0000
Mathematics	87.3333
Physics	76.0000